JAMES PATTERSON

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PROFESSIONAL SUMMARY

Data scientist with one year of experience building machine learning models and analyzing complex datasets. I started my career working with Python and SQL to solve real business problems at fintech and tech companies. I'm particularly interested in predictive modeling and helping non-technical teams understand data insights. I'm looking to deepen my expertise in deep learning and contribute to projects that have measurable business impact.

TECHNICAL SKILLS

Languages: Python, SQL, R, JavaScript

Data Analysis: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn

Machine Learning: Linear regression, logistic regression, random forests, XGBoost, neural networks

Databases: PostgreSQL, MySQL, MongoDB

Tools and Platforms: Jupyter Notebook, Git, AWS S3, Tableau, Google Sheets

Statistics: Hypothesis testing, A/B testing, correlation analysis, probability distributions

WORK EXPERIENCE

Data Scientist

DataFlow Analytics

Leeds, United Kingdom

www.dataflowanalytics.co

March 2023 - Present

Developed and deployed a customer churn prediction model using XGBoost that identified at-risk customers with 78% accuracy, resulting in targeted retention campaigns for 500+ accounts.

Automated weekly reporting by building Python scripts that pulled data from PostgreSQL databases and generated Tableau dashboards, saving the team approximately 8 hours per week.

Conducted exploratory data analysis on 2 million customer transactions to identify spending patterns and segmented customers into five behavioral groups for targeted marketing.

Collaborated with product managers and business analysts to translate business questions into data-driven solutions and presented findings using clear visualizations and written summaries.

Junior Machine Learning Engineer

TechVenture Solutions

Manchester, United Kingdom

www.techventuresolutions.io

September 2022 - February 2023

Built and trained three machine learning models to predict user subscription renewal rates using historical behavioral data and feature engineering techniques.

Created data validation pipelines in Python to ensure data quality before model training, reducing errors by approximately 40% and improving model reliability.

Assisted senior engineers in optimizing model performance by experimenting with different algorithms and hyperparameters, achieving a 12% improvement in F1 scores.

Documented all model development processes and created technical guides for junior team members to follow established workflows and best practices.

Data Analyst Intern

Yorkshire Digital Agency

Leeds, United Kingdom

June 2022 - August 2022

Analyzed website traffic data for 15 client campaigns using Google Analytics and SQL queries to identify conversion bottlenecks and recommend improvements.

Created visualizations and summary reports highlighting key performance metrics that were presented to clients in monthly business review meetings.

Cleaned and preprocessed raw data from multiple sources using Python Pandas, resolving inconsistencies and preparing datasets for analysis.

Supported the analytics team by maintaining documentation of data sources, database schemas, and analysis methodologies used across ongoing projects.

EDUCATION

BSc Data Science

University of Leeds

Leeds, United Kingdom

September 2019 - June 2022

Grade: 2:1

Relevant Coursework: Machine Learning, Statistical Analysis, Database Systems, Data Visualization, Advanced Python Programming

NOTABLE PROJECTS

Customer Segmentation Analysis

Built an unsupervised clustering model using K-means to segment 50,000 customers into distinct groups based on purchase behavior, resulting in actionable insights for marketing strategy.

Repository: github.com/jamespatterson92/customer-segmentation

Time Series Forecasting Dashboard

Developed a web application using Python and Streamlit that forecasts monthly sales using ARIMA models with interactive visualizations for stakeholder presentations.

Repository: github.com/jamespatterson92/sales-forecasting-app

House Price Prediction Model

Created a regression model predicting UK house prices with 89% accuracy using features like location, size, and market conditions, deployed as an API.

Repository: github.com/jamespatterson92/house-price-predictor

Sentiment Analysis Tool

Built a text classification model analyzing customer reviews using natural language processing techniques to automatically categorize feedback as positive, negative, or neutral.

Repository: github.com/jamespatterson92/sentiment-analyzer

CERTIFICATIONS

Google Data Analytics Professional Certificate

Completed December 2022